



Risk and Logistics Assignment 1: Tartan Trade

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February 2025

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1 Introduction

This case study considers the strategic design of a multi-echelon distribution network for Tartan Trade over a multi-year planning horizon. The company aims to capture 5% of the Scottish market. However, it is unrealistic to assume this target will be reached in the short term. Customer growth is expected to grow gradually over time and is subject to significant uncertainty. Designing warehouse infrastructure based on a single, aggregated time period may lead to inefficient outcomes such as underutilised facilities or insufficient capacity.

To address this, the multi-echelon capacitated warehouse location problem (MECWLP) is extended to include multiple time periods, allowing warehouses to be constructed gradually as demand increases. Customer demand is modelled as time dependent and uncertainty is incorporated through demand scenarios to assess the robustness of network designs. The resulting optimisation models aim to support cost-effective and flexible long-term supply chain planning.

The supply chain considered in this case study consists of three echelons: suppliers, warehouses, and customers. Suppliers provide product groups that are transported to a set of candidate warehouse locations. Warehouses serve as intermediate distribution facilities, storing products and shipping them to customers to satisfy demand. Each warehouse location can be constructed at most once and once built, remains operational for the remainder of the planning horizon.

The model assumes a single-stage distribution structure in which all customer demand is met through warehouses and direct shipments from suppliers to customers are not allowed so all flows follow the path supplier $\rightarrow$ warehouse $\rightarrow$ customer.

In the deterministic version of the model, demand values for each period are assumed to be known with certainty, providing a baseline solution and allowing analysis of warehouse construction and capacity utilisation over time. Later, uncertainty is introduced through multiple demand scenarios to assess how robust the network design is under varying market conditions.

2 Aggregation

2.1 Rationale for Aggregating Customers

Running the original problem with original number of customers, candidate locations, and suppliers caused a memory problem on our laptops. And realistically, there is no practical purpose to incorporating every single point in our model. Thus, we employed k means clustering on customer locations as well as candidate warehouse locations to simplify the model. Using three different cases of warehouse and customer location clusters, we ran the deterministic model for an hour to observe the following MIP Gap results:

Number of customer clusters	Number of Warehouse Clusters	MIP Gap after 1 hour
65	80	23.91%
80	100	33.85%
90	110	43.31%

2.1.1 Customer Aggregation

The first step for clustering was the clustering of locations of customers. In the given dataset, we had 440 locations for customers with each customer having a specific demand for a product at every period. The quantity of demand of a product in a period becomes a decision variable in our models and with 440 locations, the number of decision variables for customers become extremely large.

We intended on keeping the MIP Gap in the range of 20 – 30%, leading to the choice of customer clusters to be 65.

Each customer given had a demand associated with each product, each time period and each scenario (in the stochastic model). Once customers were clustered, the demand for each product in each period under each scenario for every cluster was the cumulative demand of all customers in the cluster for the respective product in the time period and scenario. As distance calculations were done using road distances, the closest customer location to the center of the cluster was used to calculate the approximate road distance between each customer cluster and warehouse.

2.1.2 Warehouse Clustering

By clustering the 440 candidate warehouse locations, we obtained clusters with approximately 3 or 4 candidate warehouse locations per cluster. As we intended to keep the MIP Gap in the range of 20 – 30%, we chose the number of candidate warehouse clusters to be 80.

Each warehouse location came associated with a capacity, a set up cost and an operational cost. To understand the respective values for the clusters, we utilised the candidate warehouse location that was the closest to the center of the cluster, for every cluster. Thus, if this cluster were to be chosen to be opened, the candidate warehouse location closest to the center would be the one to open, with its respective values for capacity, set up cost and operating cost remaining the same. The distance of warehouse cluster to customers and suppliers was calculated using the coordinates of the candidate warehouse nearest to the cluster center.

2.2 Elbow Plots and Average Distance to Centroid

Customer and warehouse aggregation was performed using K-means clustering on the coordinates of postcode district, and its Easting and Northing coordinates represented each customer. The elbow plots and silhouette plots can also help us understand the quality of the clustering. The elbow plot (showing within-cluster variance) is used to identify the point at which increasing the number of clusters leads to diminishing returns in terms of variance explained, while the silhouette score measures how well separated and internally coherent the resulting clusters are. Together, these plots help select a value of k that balances model simplicity with clustering quality.

The inertias for 65 customer clusters and 80 warehouse clusters are 54736544664.43 and 40573160285.31, respectively. We can use this data to calculate the average standard deviation of the distance between each point and its cluster centroid. We can see that on average, customers are 11.1 KMs away from their cluster centroid, and warehouses are 9.6 KMs away from their cluster centroid. We think that this is a suitable tradeoff considering the locations are spread out through a whole country (Scotland). Based on these diagnostics, choosing 65 clusters for customers and 80 clusters for warehouses seemed reasonable. For each cluster, the centroid location was computed as the mean of the coordinates of the customers/ warehouses belonging to that cluster.

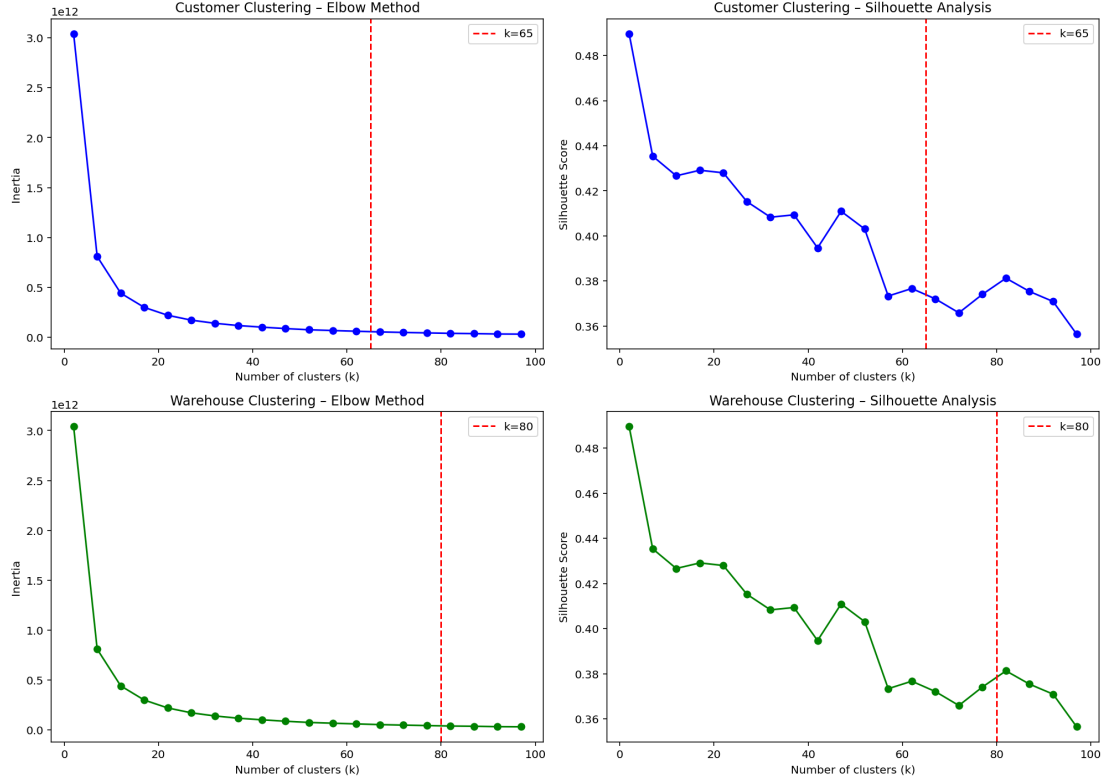


Figure 1: Elbow Plots and Silhouette Plots

3 Deterministic Multi-Period MECWLP

The deterministic multi-period MECWLP extends the classical capacitated warehouse location problem by introducing multiple time periods and inventory dynamics. The model jointly decides when and where to build warehouses, how much product to ship between each stage of the supply chain, and how much inventory to hold at each warehouse in each period. The objective is to minimise the total cost over the planning horizon, including warehouse construction costs, operating costs, and transportation costs from suppliers to warehouses and from warehouses to customers.

Sets and Indices

- I : set of customers, indexed by i
- J : set of candidate warehouse locations, indexed by j
- P : set of product groups, indexed by p
- S : set of suppliers, indexed by s
- S_p : set of suppliers for product group p , for each $p \in P$
- $T = \{1, \dots, 10\}$: set of time periods (years), indexed by t
- R : Different scenarios of customer demand considered, indexed by r

Parameters

- b_{ipt} : demand of customer i for product p in period t
- f_j : construction cost of building a warehouse at candidate location j
- g_j : annual operating cost of a warehouse at candidate location j
- K_j : capacity of a warehouse at candidate location j
- K'_s : capacity of supplier s

c_{ji} : unit transportation cost (per quantity of product) from warehouse at location j to customer i
 c'_{sj} : unit transportation cost (per quantity of product) from supplier s to warehouse at location j
 q_r : probability of scenario r occurring, i.e. $\frac{1}{|R|}$ (since each scenario is equally likely)

Decision Variables

$y_{jt} \in \{0, 1\} \quad \forall j \in J, t \in T$: 1 if a warehouse is constructed at candidate location j in period t , 0 otherwise
 $z_{jt} \in \{0, 1\} \quad \forall j \in J, t \in T$: 1 if a warehouse is operational in candidate location j in period t , 0 otherwise
 $x_{ijpt} \geq 0 \quad \forall i \in I, j \in J, p \in P, t \in T$: quantity of product p supplied from warehouse at location j to customer i at time period t
 $w_{sjt} \geq 0 \quad \forall s \in S, j \in J, t \in T$: quantity supplied of product p from supplier s to warehouse at location j at time period t

Objective Function

The objective function minimises total construction, operating and transportation costs over the planning time. The transportation costs include transportation from the warehouses to customers, and from the suppliers to warehouses. It is assumed transportation costs are linear and proportional to supplied quantities, and inventory holding costs are neglected; inventory is only constrained by capacity.

$$\min \sum_{j,t} f_j y_{jt} + \sum_{j,t} g_j z_{jt} + \sum_{i,j,p,t} c_{ji} x_{ijpt} + \sum_{s,j,t} c'_{sj} w_{sjt}$$

Constraints

Total supply from all warehouses must exactly meet customer demand for each period:

$$\sum_{j \in J} x_{ijpt} = b_{ipt} \quad \forall i \in I, p \in P, t \in T$$

Each warehouse can be constructed at most once over the planning horizon:

$$\sum_{t \in T} y_{jt} \leq 1 \quad \forall j \in J$$

A warehouse is operational only after it is constructed:

$$z_{jt} = \sum_{r \leq t} y_{jr} \quad \forall j \in J, t \in T.$$

All the products received at each period at each warehouse will be sent to customers at the same period:

$$\sum_{s \in S_p} w_{sjt} = \sum_{i \in I} x_{ijpt} \quad \forall j \in J, t \in T, p \in P$$

Warehouse shipments in each period cannot exceed its capacity and occur only if operational:

$$\sum_{i,p} x_{ijpt} \leq K_j z_{jt} \quad \forall j \in J, t \in T.$$

Supplier shipments in each period cannot exceed their production capacity:

$$\sum_{j \in J} w_{sjt} \leq K'_s \quad \forall s \in S, t \in T.$$

In the proposed formulation, there is no explicit constraint linking the warehouse operational variable z_{jt} directly to the inbound flows w_{sjt} . This is intentional and does not affect model feasibility. The linkage between warehouse availability and inbound shipments is implicitly enforced through the outbound warehouse capacity constraints.

4 Results

The deterministic multi-period MEWCWLP was implemented in Python using FICO Xpress. The problem was solved using branch and bound with cutting planes. The LP relaxation was solved in approximately 6 seconds. The full MIP search terminated after 291.51 seconds due to the relative MIP gap stopping criterion of 24%. The final solver statistics are:

- MIP Gap: 23.91%
- Solution time: 291.51 seconds
- Solutions found: 8
- Expected Total Cost: £39,865,247.63

The solver stopped once the relative gap fell below the specified threshold of 24%. Therefore, optimality was not proven, but a feasible solution within 23.91% of the best known bound was obtained.

Table 1: Cost Summary

Cost Component	Value
Total Construction Cost	£11,544,000.00
Total Operation Cost	£8,673,700.00
Warehouse → Customer Transportation	£15,867,936.36
Supplier → Warehouse Transportation	£3,779,611.27

Warehouse-to-customer transportation represents the largest portion of the total cost. Fixed costs (setup and operating) are also substantial, reflecting significant investment in warehouse infrastructure.

Table 2 presents the optimal warehouse construction schedule for the deterministic model. A total of 12 warehouses are opened over the planning horizon, visualised in Figure 2. The solution exhibits significant front-loading, with five warehouses opened in period 1. This indicates that the model invests heavily in initial capacity to satisfy predicted demand levels.

Table 2: Optimal Warehouse Opening Schedule

Period	Warehouse ID
1	42
1	58
1	61
1	63
1	76
2	36
3	68
4	6
4	31
6	78
6	79
9	71

Additional warehouses are opened gradually in later periods reflecting incremental expansion as demand increases over time. Notably, no warehouses are opened in periods 5, 7, or 8, suggesting that existing capacity is sufficient during those stages.

Since the deterministic model assumes demand forecasts are exact, warehouse opening decisions are made strictly according to expected growth. This leads to substantial early investment in infrastructure to minimise transportation costs over the full planning horizon.

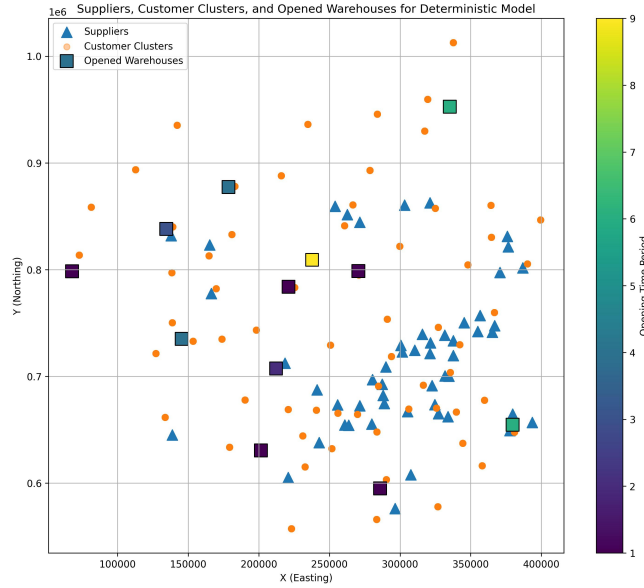


Figure 2: Deterministic Model Visualisation

5 Stochastic MECWLP with Demand Uncertainty

The sets, indices, parameters and decision variables are the same as those in the Deterministic Multi-Period MECWLP. The stochastic model extends the deterministic model by incorporating demand uncertainty through scenarios which are added to the relevant decision variables described below.

Extended Decision Variables

$x_{ijptr} \geq 0$: Quantity of product p supplied from warehouse at location j to customer i at time period t under scenario r

$w_{sjtr} \geq 0$: Quantity supplied of product p from supplier s to warehouse at location j at time period t under scenario r

Extended Parameters

b_{iptr} : demand of customer i for product p in period t in scenario r

Objective function

The costs incurred in the process are given by

- Fixed costs for warehouses = $\sum_{j \in J} \sum_{t \in T} f_j y_{jt}$
- Operating costs for all open warehouses = $\sum_{j \in J} \sum_{t \in T} g_j z_{jt}$
- Total cost of transporting from suppliers to open warehouses (considering all scenarios) = $\sum_{r \in R} q_r (\sum_{s \in S} \sum_{j \in J} \sum_{t \in T} c'_{sj} w_{sjtr})$
- Total cost of transporting from warehouses to customers (considering all scenarios) = $\sum_{r \in R} q_r (\sum_{i \in I} \sum_{j \in J} \sum_{p \in P} \sum_{t \in T} c_{ji} x_{ijptr})$

Thus, as we intend to minimize total cost, we have the objective function given by,

$$\min \left[\sum_{j \in J} \sum_{t \in T} f_j y_{jt} + \sum_{j \in J} \sum_{t \in T} g_j z_{jt} + \sum_{r \in R} q_r \left(\sum_{s \in S} \sum_{j \in J} \sum_{t \in T} c'_{sj} w_{sjtr} \right) + \sum_{r \in R} q_r \left(\sum_{i \in I} \sum_{j \in J} \sum_{p \in P} \sum_{t \in T} c_{ji} x_{ijptr} \right) \right]$$

Constraints

Satisfying customer demand in every scenario:

$$\sum_{j \in J} x_{ijptr} = b_{iptr} \quad \forall i \in I, \forall p \in P, \forall t \in T, \forall r \in R$$

Amount of total product supplied from a warehouse does not exceed the capacity it has (provided it is open):

$$\sum_{i \in I} \sum_{p \in P} x_{ijptr} \leq K_j z_{jt} \quad \forall j \in J, \forall r \in R, \forall t \in T$$

Any warehouse can either be not built or built only once:

$$\sum_{t \in T} y_{jt} \leq 1 \quad \forall j \in J$$

Once a warehouse is built, it is operational till the end of the time periods:

$$z_{jt} = \sum_{\tau=1}^t y_{j\tau} \quad \forall j \in J, \forall t \in T$$

All the products received at each period at each warehouse will be sent to customers at the same period under any scenario:

$$\sum_{s \in S_p} w_{sjtr} = \sum_{i \in I} x_{ijptr} \quad \forall j \in J, t \in T, p \in P, r \in R.$$

Amount of product supplied by supplier should not exceed the capacity the supplier has at any time period:

$$\sum_{j \in J} w_{sjtr} \leq K'_s \quad \forall s \in S, \forall t \in T, \forall r \in R$$

Domain constraints:

- $x_{ijptr}, w_{sjtr} \in \mathbb{R}_0^+$
- $y_{jt}, z_{jt} \in \{0, 1\}$

6 Results (Stochastic Case)

The Stochastic MEWCWLP with Demand Uncertainty was implemented in Python using FICO Xpress. The problem was solved using branch and bound with cutting planes. The solver statistics are summarised below:

- MIP Gap: 0.00%
- Solution time: 578.93 seconds
- Nodes explored: 3
- Solutions found: 5
- Expected Total Cost: £31,974,650.37

The solver reached proven optimality with a negligible optimality gap.

Table 3: Cost Summary

Cost Component	Value
Total Construction Cost	£8,344,000.00
Total Operation Cost	£6,826,800.00
Warehouse → Customer Transportation	£14,407,838.98
Supplier → Warehouse Transportation	£2,396,011.38

Transportation from warehouses to customers represents the largest share of total cost, accounting for approximately 45% of the total. Setup and operating costs together account for approximately 47%. Supplier to warehouse transportation represents a relatively small proportion.

The optimal solution determines both the location and timing of warehouse construction. Since actual demand in period t can deviate by up to $\pm 0.1t$ from predicted demand, capacity decisions must hedge against higher-demand realisations, particularly in later periods. As a result, facilities are opened conservatively to guarantee feasibility for every scenario.

Table 4: Optimal Warehouse Opening Schedule under Demand Uncertainty

Period	Warehouse ID
1	9
1	28
1	42
1	63
2	13
2	20
2	44
3	61
4	34
4	71
5	58
8	75

Table 4 shows the optimal warehouse construction schedule. A total of 12 warehouses are opened over the planning horizon, visualised in Figure 3. Four warehouses are opened in period 1, indicating that capacity is front-loaded to hedge against increasing demand uncertainty in later periods. Subsequent openings occur gradually, with additional facilities added in periods 2–5 and a final warehouse opened in period 8 to accommodate long-term demand growth.

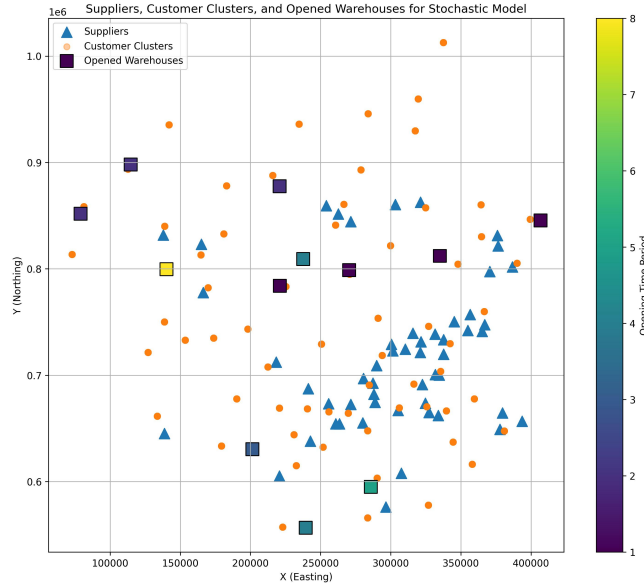


Figure 3: Stochastic Model Visualisation

7 Less conservative Stochastic formulation

In real logistics systems, it is often acceptable to either leave some demand unmet in rare cases or temporarily exceed capacity limits. We propose the following less conservative formulations.

7.1 Allowing Unmet Demand with Penalty

We introduce a non-negative slack variable:

$$e_{iptr} \geq 0 \quad : \text{unmet demand of customer } i \text{ for product } p \text{ at time } t \text{ under scenario } r$$

The demand satisfaction constraint becomes:

$$\sum_{j \in J} x_{ijptr} + e_{iptr} = b_{iptr} \quad \forall i \in I, \forall p \in P, \forall t \in T, \forall r \in R$$

A penalty cost for unmet demand is added to the objective function, which is given by:

$$\sum_{r \in R} q_r \sum_{i \in I} \sum_{p \in P} \sum_{t \in T} \pi_{ip} e_{iptr}$$

where π_{ip} is a penalty parameter reflecting the cost of lost sales for customer cluster i and product p . This parameter is allowed to depend on the customer cluster, since the number of customers represented by each cluster affects the severity of lost demand; failing to serve a large cluster leads to higher economic and service losses than failing to serve a small cluster. Additionally, different product groups may have different urgency levels or strategic importance, which is represented through product-specific penalty values.

This formulation allows limited unmet demand in extreme scenarios, but only when the cost of opening additional warehouses outweighs the penalty.

7.2 Chance-Constrained Formulation

Let b_{it} denote the predicted demand of customer i in period t . The actual demand $\hat{b}_{it}$ is assumed to lie in the interval

$$\hat{b}_{it} \in [(1 - 0.1t) b_{it}, (1 + 0.1t) b_{it}],$$

reflecting a maximum deviation of $\pm 10\%$ per period.

Given a service level requirement $1 - \alpha$, we aim to ensure that customer demand is satisfied with probability at least $1 - \alpha$. This can be expressed through the chance constraint

$$\mathbb{P} \left(\sum_{j \in J} x_{ijpt} \geq \hat{b}_{it} \right) \geq 1 - \alpha, \quad \forall i \in I, p \in P, t \in T.$$

Since the demand distribution is known, this probabilistic constraint can be transformed into a deterministic equivalent by using the $(1 - \alpha)$ -quantile of the demand distribution. Let F_{it} denote the cumulative distribution function (CDF) of $\hat{b}_{it}$. We define the critical demand level b_{it}^{crit} as

$$b_{it}^{\text{crit}} = F_{it}^{-1}(1 - \alpha),$$

which represents the smallest demand value such that the probability of actual demand exceeding this level is at most α , i.e.

$$\mathbb{P}(\hat{b}_{it} > b_{it}^{\text{crit}}) \leq \alpha.$$

This critical value can be easily calculated as the demand in each period follows a uniform distribution, and we can calculate the 90th percentile of each distribution with ease. The deterministic constraint is then replaced by the chance constraint, given by

$$\sum_{j \in J} x_{ijpt} \geq b_{it}^{\text{crit}}, \quad \forall i \in I, p \in P, t \in T,$$

which guarantees that demand is satisfied in at least $100(1 - \alpha)\%$ of all possible realisations.

8 Conclusion

A comparison of the deterministic and stochastic warehouse opening schedules reveals several notable differences. Both models open a total of 12 warehouses over the planning horizon. However, the timing and spatial selection of the facilities differ.

In the deterministic model, five warehouses are opened in period 1, indicating substantial front-loading of capacity based purely on forecasted demand. In contrast, the stochastic model opens four warehouses in period 1, followed by a more gradual expansion in subsequent periods. The stochastic solution therefore spreads investment more evenly over time.

This difference reflects the modelling assumptions. The deterministic model assumes demand forecasts are exact and therefore optimises capacity solely with respect to predicted growth. In contrast, the stochastic model must ensure feasibility across all demand scenarios, where demand may deviate by up to $0.1t$ in period t . As uncertainty increases over time, the stochastic model hedges against higher-demand realisations by adjusting the timing and placement of warehouse openings.

It is also important to note that the deterministic model was not solved to proven optimality (MIP gap of 23.91%), whereas the stochastic model was solved to optimality (0.00% gap). Consequently, the deterministic total cost is likely overestimated, and the cost comparison should be interpreted with caution. Nonetheless, the differences in opening schedules clearly illustrate the impact of incorporating demand uncertainty into long-term facility planning.

Overall, incorporating uncertainty allows for more robust and flexible warehouse planning, potentially reducing the risk of unmet demand and costly emergency expansions. While the stochastic approach may involve slightly higher upfront planning effort, it provides greater confidence in the system's ability to handle variability in customer demand. Future work could explore additional uncertainty factors, such as transportation delays or supplier disruptions, to further enhance decision robustness.